CHAPTER 4

# The Suction Mechanism of the Maxillary and Mandibular Complete Dentures

# 1 What is the suction mechanism of the maxillary complete denture?

## [1] Sealing mechanism in the maxilla

The biggest complaint among complete denture wearers is the dropping down of the maxillary denture, followed by the lifting of mandibular dentures. Therefore, we must achieve an effective suction of the maxillary denture first.

The sealing mechanism in the maxilla is more straightforward than in the mandible and consists of two types of closure.

### 1) Interior/exterior double closure (labiobuccal area) (Fig 4-1)

The exterior surface of the denture base in the anterior and posterior areas of the maxillary alveolar ridge contacts the labiobuccal mucosa, while the interior surface of the denture base is in contact with the alveolar mucosa. The labiobuccal border of the denture base is sandwiched between the two mobile mucosae. The author calls this sealing mechanism "interior/exterior double closure," which is the strongest form of the seal. Because of minimal displacement of the mucobuccal fold in the maxilla, the maxillary molar area is relatively easy to border-mold with compound or a heavy-body silicone impression material to obtain a good seal.

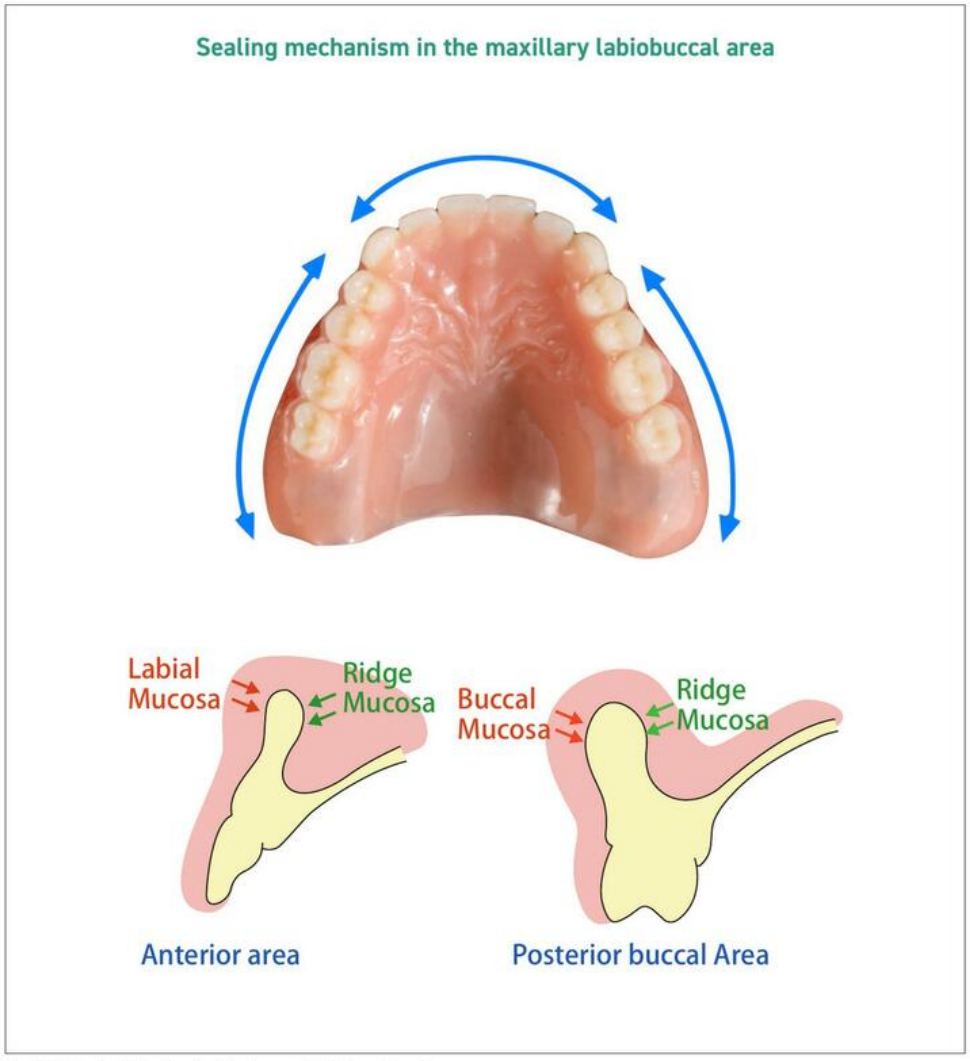



**Fig 4-1** Interior/exterior double closure (labiobuccal area).

## 2) Close-contact closure (posterior palatal area) (Fig 4-2)

The posterior palatal seal of the maxillary denture is accomplished by the contact of the interior surface of the denture base and palatal mucosa via a film of saliva. The author calls this sealing mechanism the "close-contact closure." The moment a greater dislodging force than the surface tension of saliva is applied to the denture, air leaks and the seal is easily broken in this area. This necessitates the reinforcement of the seal by taking a functional impression of the posterior palatal border area at the completion of the final impression with a relatively fluid wax, compound or silicone material (**Fig 4-3**). Carving a post dam on the cast is also an effective way to ensure a good posterior palatal seal.

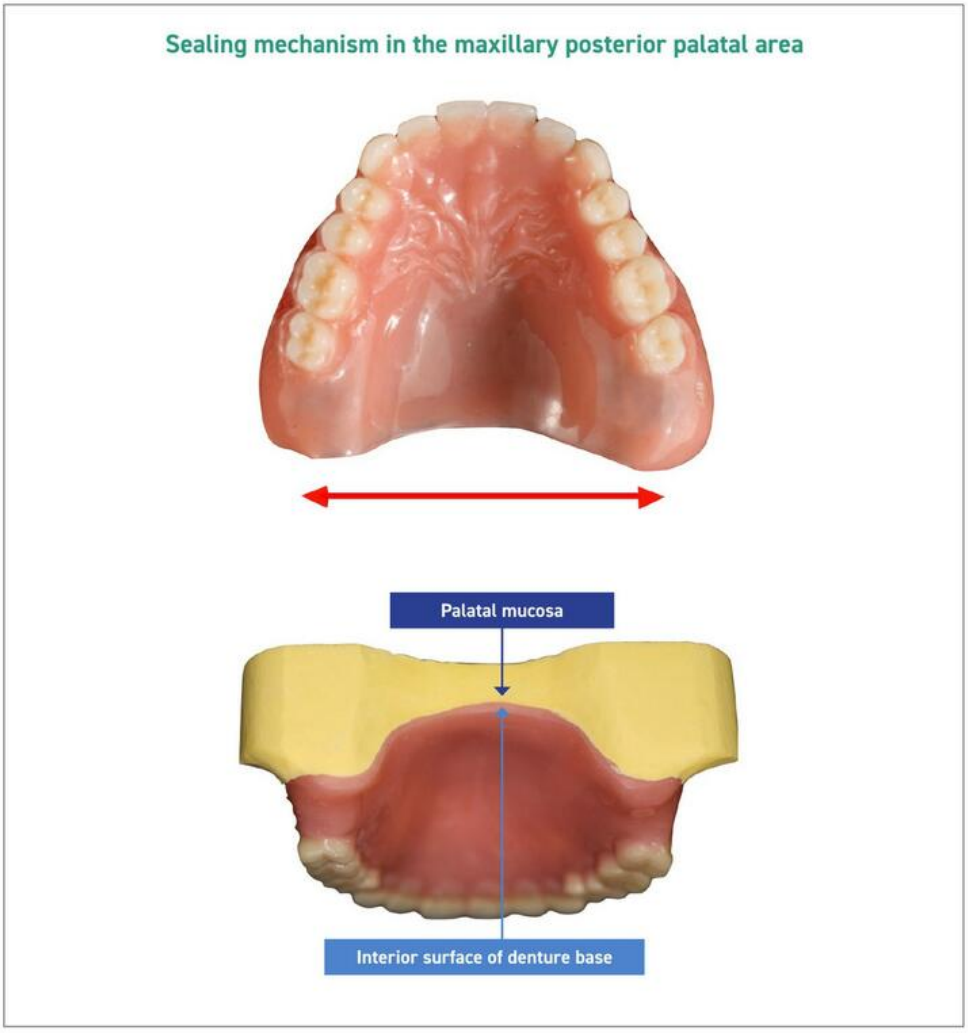



**Fig 4-2** Close contact closure (posterior palatal area).

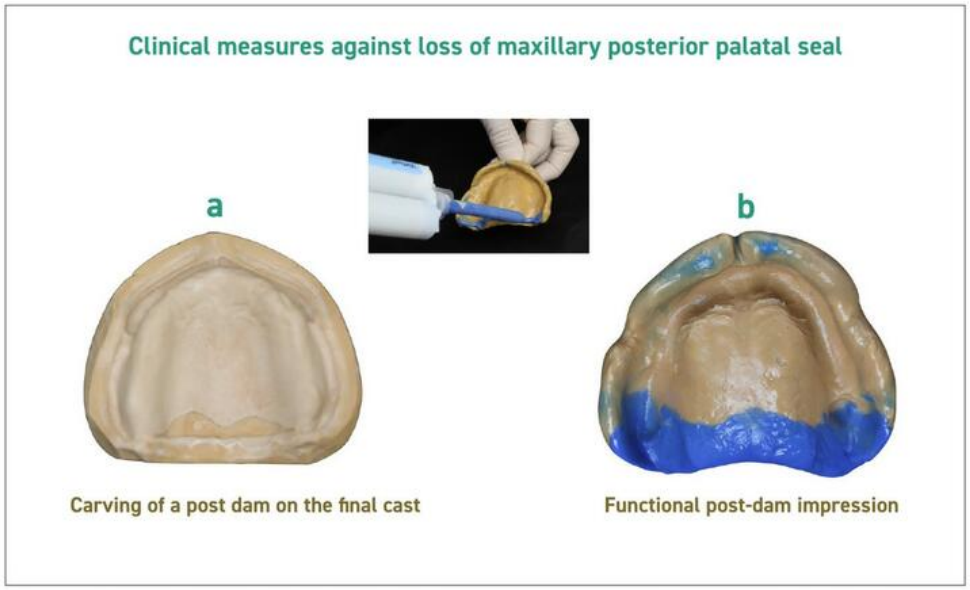



**Fig 4-3a** Grind a part of the posterior palatal area with a plaster evans carver as a preventive measure against loss of close contact closure due to impression error and/or resin polymerization shrinkage, so that a good posterior palatal seal can be obtained.
**Fig 4-3b** The posterior palatal seal can be reinforced by adding an impression material to the posterior palatal area and applying functional pressure at the end of final impression.

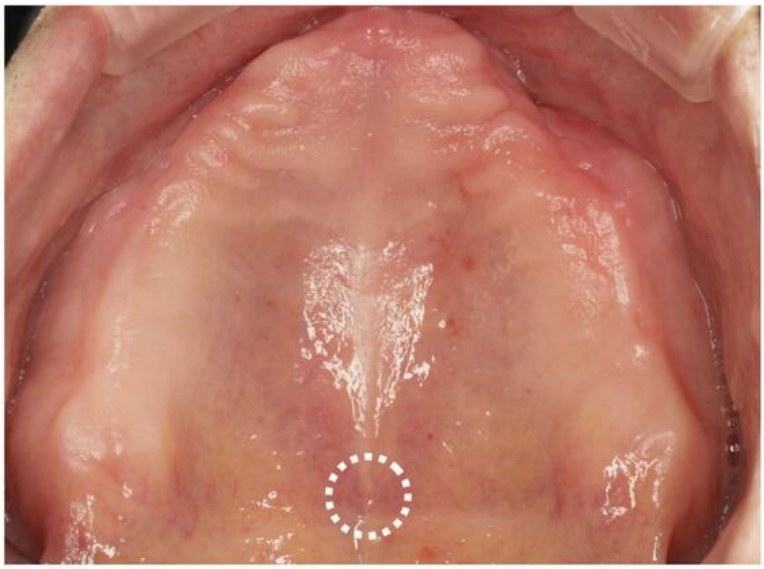


**Fig 4-3c** Extra care must be taken when notches and grooves are present in the posterior palatal area, since these irregularities may increase the risk of losing the palatal seal.

## Summary

The suction seal of the maxillary denture consists of two types of closure; interior/exterior double closure in the labiobuccal area and close-contact closure in the posterior palatal area (**Fig 4-4**). The former provides a strong seal, while the latter gives only a weak seal. Therefore, reinforcement of close-contact closure in the palatal posterior area holds the key to the achievement of suction in the maxilla.

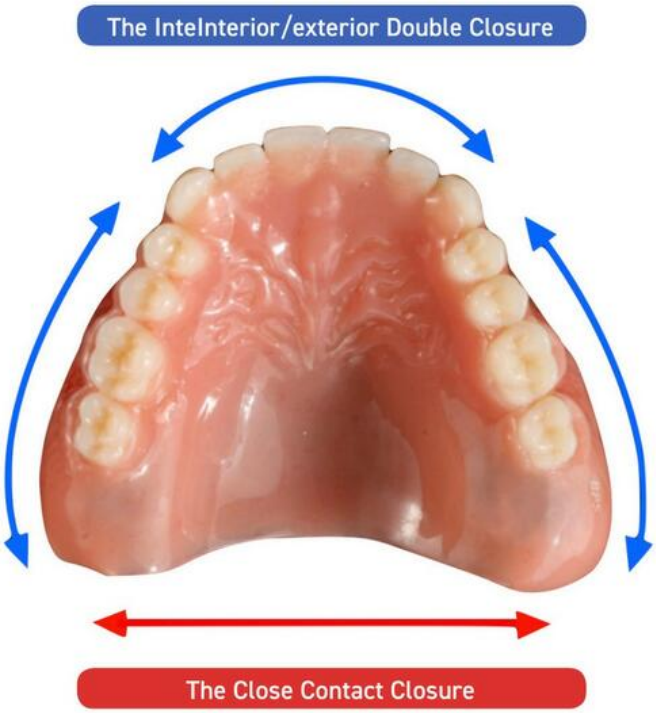



**Fig 4-4** Blue→ Interior/exterior double closure in the labiobuccal area.
Red → Close contact closure in the posterior palatal area

## 2 What is the suction mechanism of the mandibular complete denture?

### [1] Sealing mechanism of the mandibular complete denture

The sealing mechanism is more complicated in the mandible than in the maxilla. It consists of the following 4 main types of closure (**Fig 4-5**).

- Blue → Interior/exterior double closure,
- yellow → Interior/exterior double closure when spongy tissue is present in the sublingual fold region,
- green → compensatory closure in the retromylohyoid muscle fossa region and
- red → close-contact closure of the interior side of the denture base and retromolar pad tissue (interior closure) and exterior closure over the denture base with the tongue sidewall contacting the buccal mucosa (exterior closure).

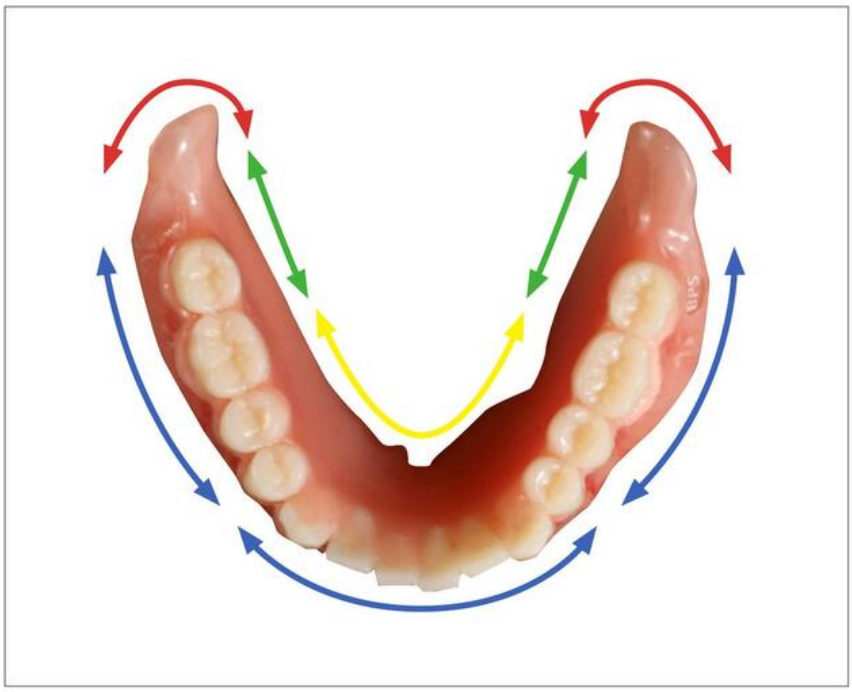


**Fig 4-5** The suction mechanism is more complicated in the mandible than in the maxilla.

**Fig 4-6** shows tongue movements during the opening and closing of the mouth. When opening, the tongue recedes, and space appears. The seal can be easily broken in the sublingual fold region and around the retromolar pads as the contact with the tongue is lost. Space tends to develop between the denture borders and tongue during tongue movement, particularly in the sublingual fold area and retromylohyoid fossa regions, requiring special attention to these regions of the mandible. We can easily imagine how one tiny spot of air leakage can break the seal, resulting in loss of suction and lifting of the denture.

The border seal in the retromolar pad region is difficult to achieve unless we completely change our traditional way of thinking. We must thoroughly understand the sealing mechanism in the retromolar pad region before embarking on mandibular denture fabrication. This section describes in detail how to take a preliminary impression using the Frame Cut Back (FCB) Tray that can capture the natural form of the retromolar pads in the mandibular rest position without deformation, how to create a good seal over the retromolar pad region with the side of the tongue and buccal mucosa, and how to outline the cast for the fabrication of the custom tray.

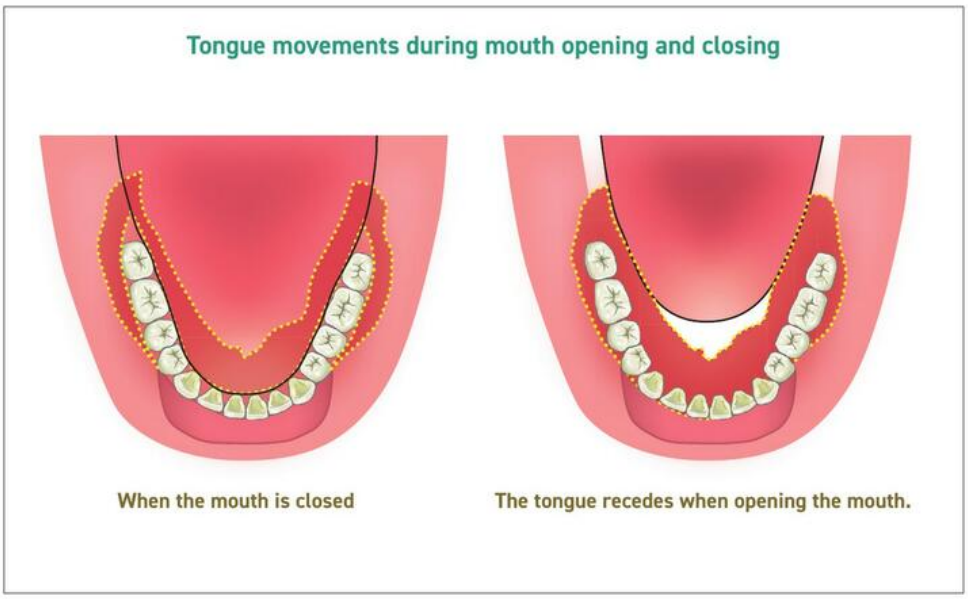



**Fig 4-6** It is difficult to obtain a good seal in areas with empty space, i.e. around the retromolar pads and sublingual folds.

## 1) Interior/exterior double closure (labiobuccal area)

A labiobuccal seal of the mandibular denture base is created when the labiobuccal mucosa contacts the exterior denture base surface, and the alveolar mucosa contacts the interior denture base surface. In other words, the denture base is sandwiched between these two mobile mucosae. The author calls this the "interior/exterior double closure," and it provides the most stable and strongest seal (**Fig 4-7**).

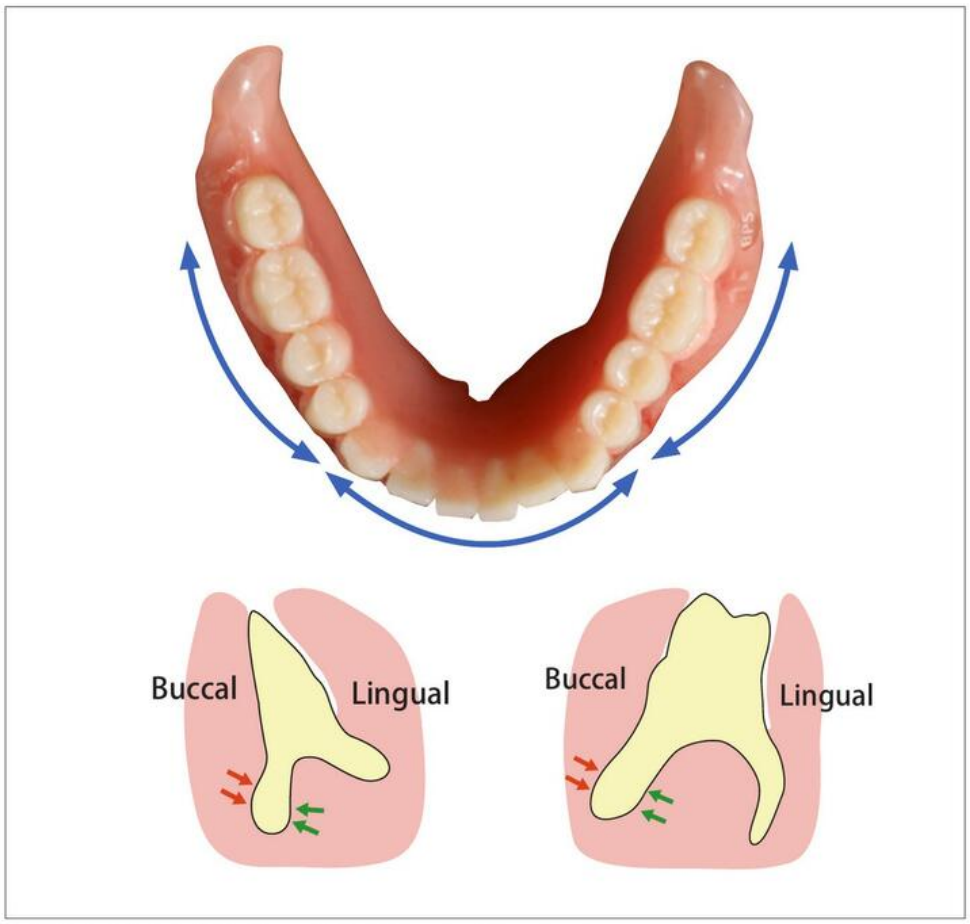



**Fig 4-7** Interior/exterior double closure with the labiobuccaland ridge mucosae.

## 2) Seal in the lingual area

The lingual seal consists of a closure in the anterior sublingual fold region and closure in the posterior retromylohyoid fossa region(**Fig 4-8**). The sealing mechanism differs between these two (chart from left to right).

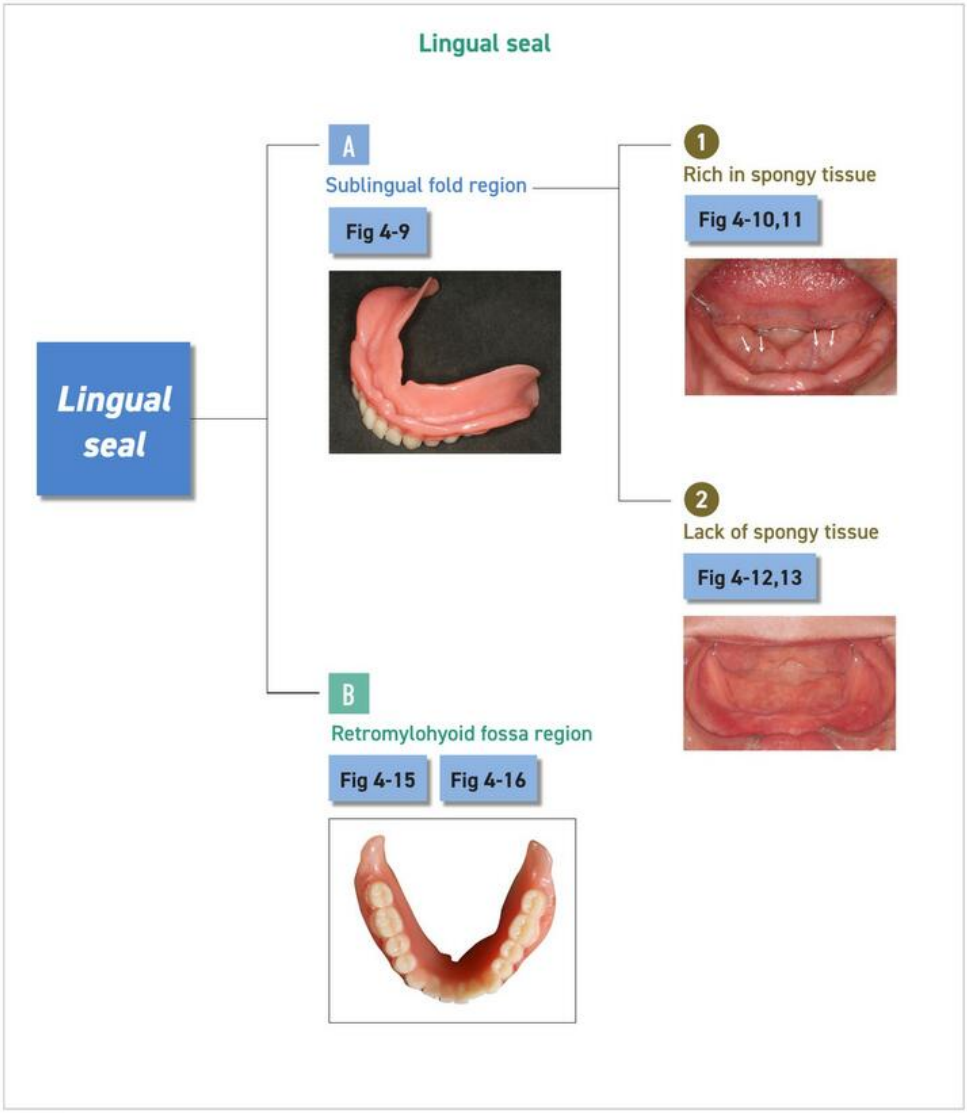



**Fig 4-8**

## A Seal in the sublingual fold region

### 1 When the region is rich in spongy tissue

This section describes the seal in the sublingual fold region (**Fig 4-9**). **Fig 4-10a** depicts a case with an abundance of spongy tissue posterior to the residual ridge. The soft spongy tissue can be stretched deep and wide during functional impression to obtain a good, thick imprint of the denture border area. This provides an extended area of contact between the denture base and mucosa; hence a strong and stable seal is created (**Fig 4-10b**).



**Fig 4-9a,b** Seal in sublingual fold region.

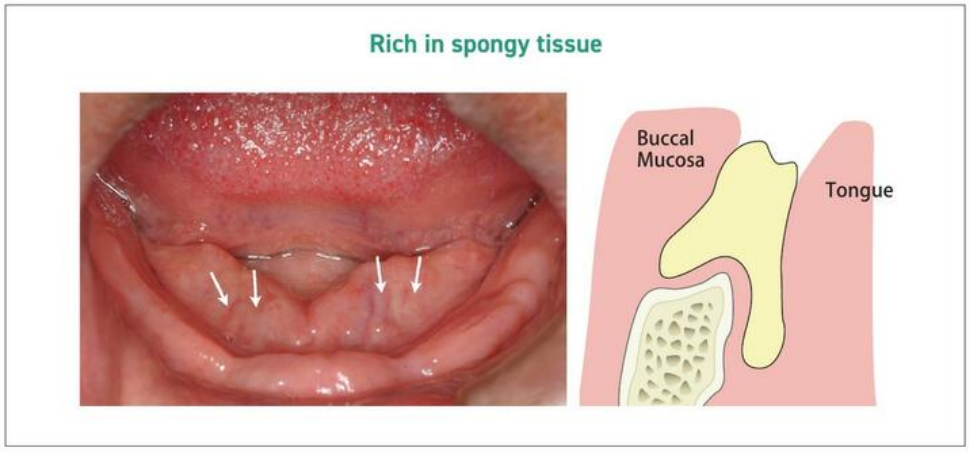



**Fig 4-10a** The sublingual fold region posterior to the alveolar ridge is rich in spongy tissue. A strong suction can be expected in this case.
**Fig 4-10b** A deep and thick border of denture base.

There is another benefit. **Fig 4-11a** illustrates the mucosal status during mandibular rest position with the blue line representing the denture base surface that contacts the sublingual mucosa. **Fig 4-11b** depicts the unchanged contact of the mucosa with the denture base during opening of the mouth. The tongue recedes during mouth opening and moves in all directions during chewing. However, the denture surface area shown in blue remains in stable contact with the sublingual mucosa. Thus, the most effective interior/exterior double closure can be expected when there is a large amount of soft spongy tissue available in the sublingual fold region.

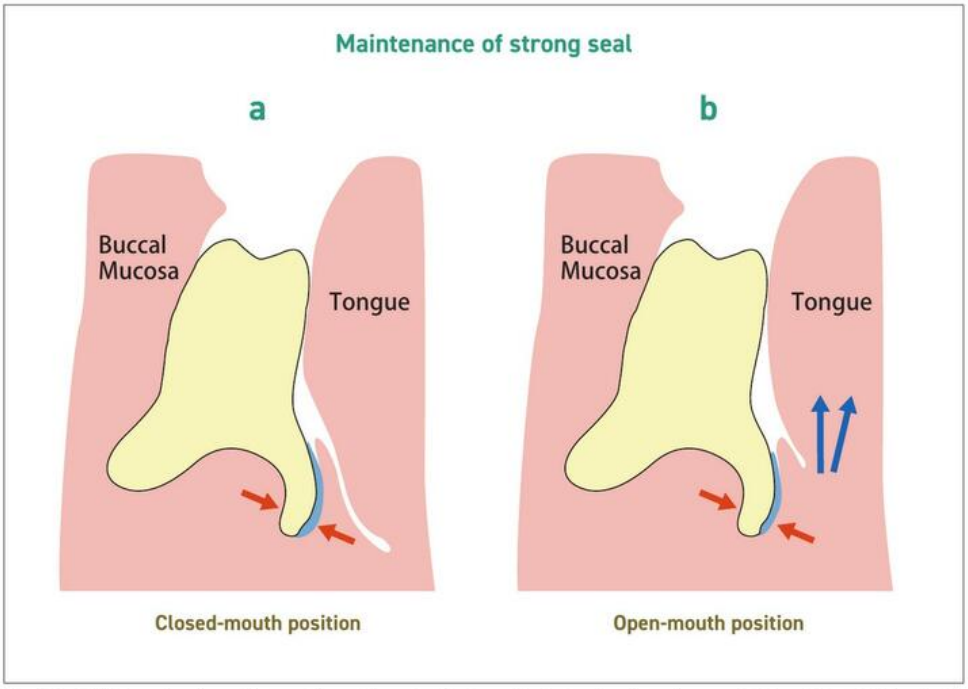



**Fig 4-11a,b** With an abundance of spongy tissue, the contact between the denture border and mucosa in the sublingual fold region is maintained even during tongue movement, providing a stable and strong seal.

## 2 When spongy tissue is lacking

When the sublingual fold region lacks spongy tissue, the suction seal is easily quickly broken as the tongue retracts during mouth opening (**Figs 4-12, 13**). This lack of spongy tissue makes it challenging to obtain suction of the mandibular complete denture. How to approach a difficult-to-treat case lacking in spongy tissue will be discussed in Chapter 6.

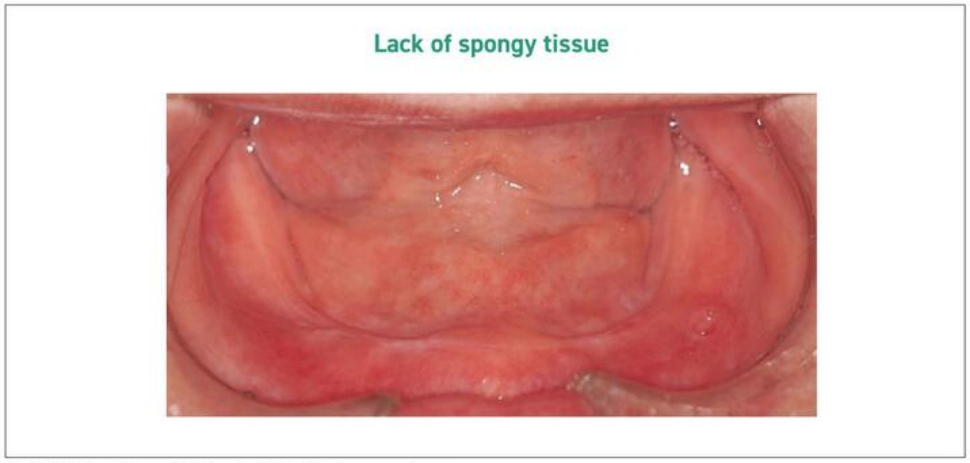



**Fig 4-12** The sublingual fold region is lacking in spongy tissue.

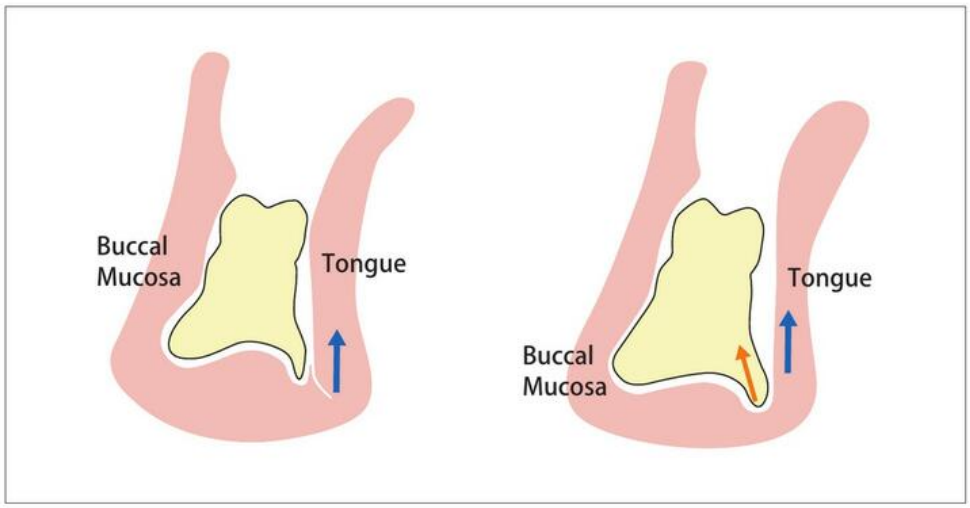



**Fig 4-13a** Transition from closed- to open-mouth position
**Fig 4-13b** Because of a limited contact between the denture border and mucosa in the sublingual fold region, the denture lifts with the slightest tongue movement.

## 3 The oral environment of the sublingual fold region in the mouth and on the cast

Since the impression accurately represents the oral environment, it is essential to correlate intraoral observations with those found on the dental cast (**Fig 4-14**). A comparison of cases abundant in spongy tissue and those lacking in spongy tissue provides dental technicians with relevant information for their laboratory work. A deep and thick lingual fold, as it appears on the cast, gives an advantage in achieving good mandibular denture suction and seal. Conversely, a shallow lingual groove is indicative of an adverse condition for suction with a lack of spongy tissue in the sublingual fold region.

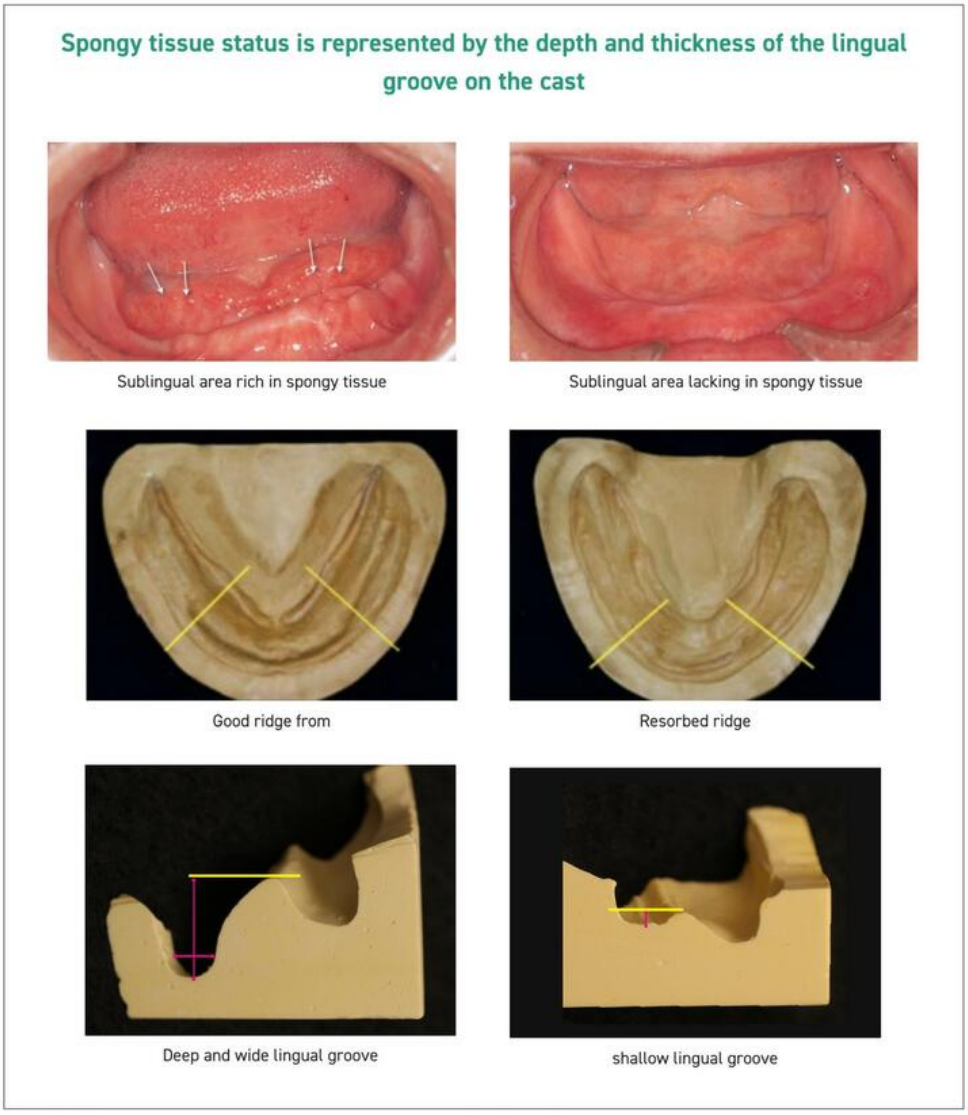



**Fig 4-14** Spongy tissue status is represented by the depth and thickness of the lingual groove on the cast.

## B Compensatory closure in the retromylohyoid fossa region (Figs 4-15, 16)

There is a space below the denture border in the retromylohyoid fossa region, as shown in **Fig 4-17**. A part of the lingual denture border is out of contact with the sublingual mucosa in the retromylohyoid fossa region. Therefore, the seal in this region is in a strict sense incomplete. In this region, a seal is maintained with the side of the tongue pressing against the polished lingual surface of the denture base, rather than the intimate contact between the denture base and oral mucosa. The author calls this type of seal "compensatory closure." The presence of space below the lingual denture border justifies the extension of the denture border deeper down into the retromylohyoid fossa region if the operator so wishes.

To achieve this compensatory closure, a wall must be created to resist applied tongue pressure when the root of the tongue presses against the polished lingual surface (**Fig 4-18**). To accomplish this objective, the custom tray border needs to be extended at least 2 to 3 mm beyond the mylohyoid ridge, regardless of the alveolar ridge condition. If the denture base is short of the mylohyoid ridge without adequate extension into the retromylohyoid fossa region, as shown in **Fig 4-19**, the tongue displaces the denture sideways. This is particularly important in patients with resorbed alveolar ridges as air leakage easily breaks the suction seal.

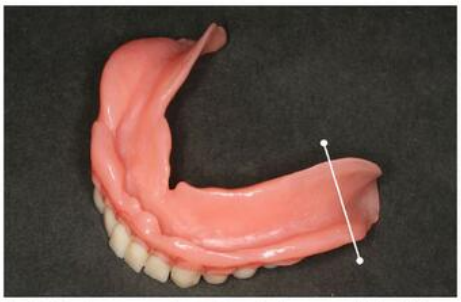


**Fig 4-15** A cross-sectional view of the retromylohyoid fossa region.



**Fig 4-16** Compensatory closure in the retromylohyoid fossa region.

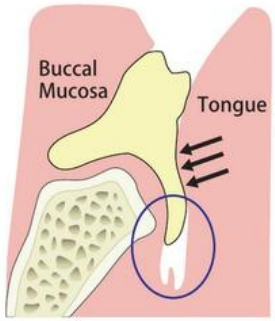



**Fig 4-17** Empty space is present below the lingual denture border.

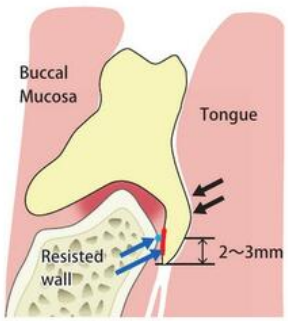



**Fig 4-18** A wall to resist tongue pressure is required to obtain compensatory closure.

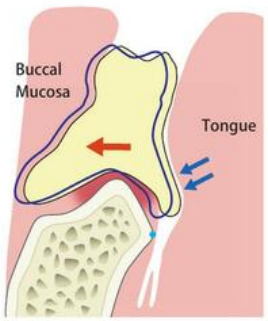



**Fig 4-19** Seal is incomplete without the wall to resist tongue pressure, resulting in loss of suction.

However, some previous reports suggested that the denture borders should not be extended beyond the mylohyoid ridge. Nagel [12] reported from a myologic viewpoint in 1958 that the extension of the denture base into the retromylohyoid fossa region interfered with muscle movements (**Fig 4-20**). Others have expressed physiological views against the extension of the denture border into the retromylohyoid muscle region, on the basis that the space in this region has been present since before a patient's edentulism. The mylohyoid fossa area is therefore physiologically indispensable and should not be violated. Regarding the myologic issue, however, modern anatomists maintain that there is no problem with some extension of the denture border into the retromylohyoid fossa region since the mylohyoid muscle fibers are oriented downward even when the muscle is under strain (**Fig 4-21**).

What about other physiological issues? According to the principle of sealing the entire denture borders to achieve mandibular denture suction, compensatory closure in the retromylohyoid fossa region is essential. To accomplish this closure, it is necessary to extend the denture base downward past the mylohyoid ridge. This extension is an appropriate compromise. Just like a maxillary denture covers the entire surface of the non-tooth-bearing palatal mucosa to increase the area of contact between the interior surface of the denture base and the mucosa for denture retention, the mandibular denture base extension into the retromylohyoid fossae region can be regarded as a necessary means to facilitate lower denture suction (**Fig 4-22**).

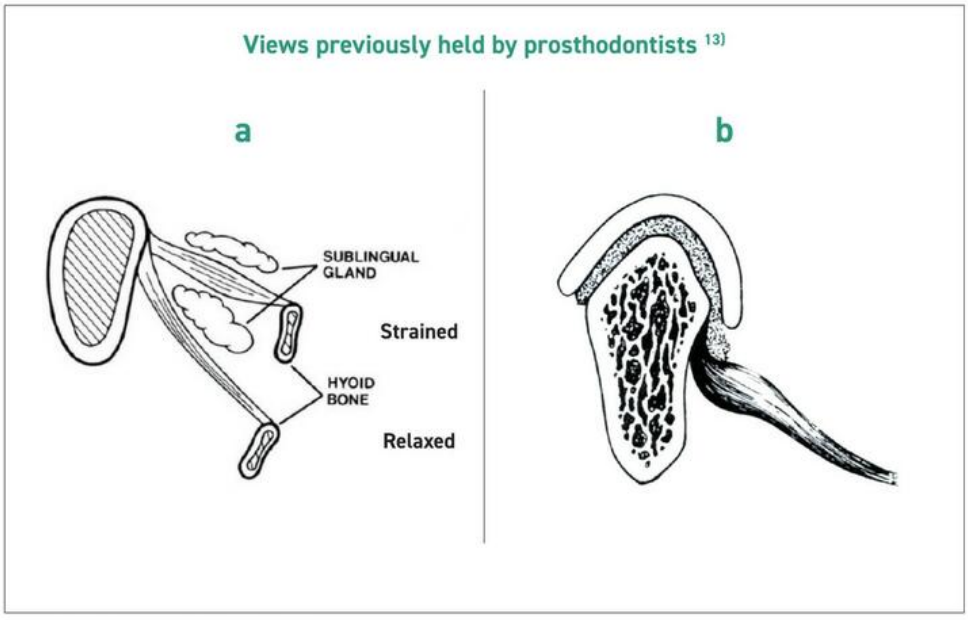



**Fig 4-20a** Mylohyoid muscle strained, relaxed.
**Fig 4-20b** Custom tray extended beyond the mylohyoid ridge interferes with muscle movement.

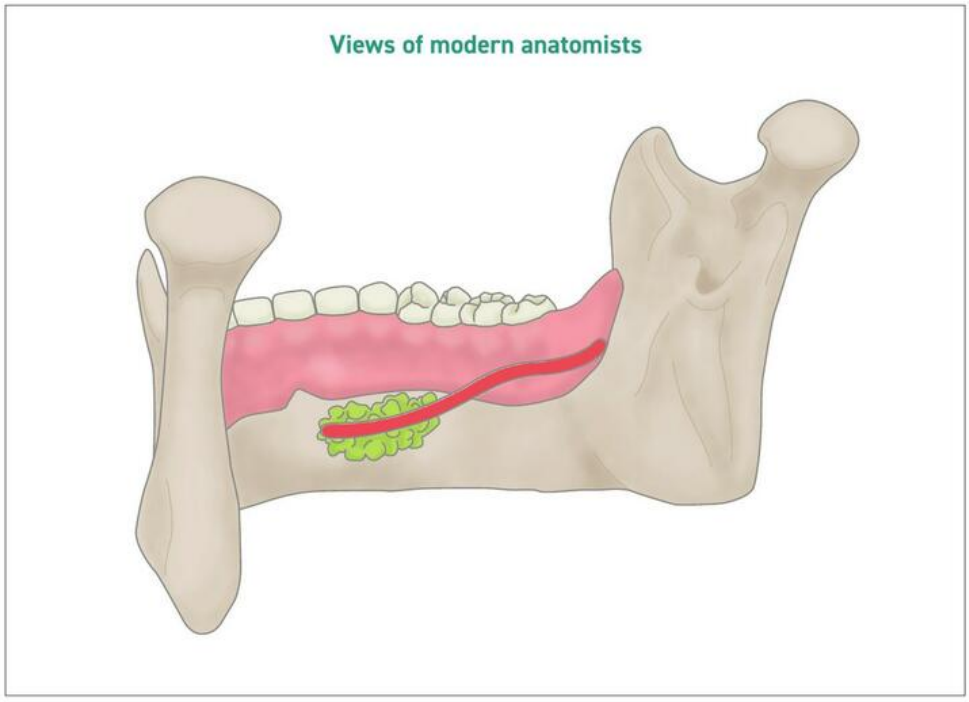



**Fig 4-21** The denture base can be extended beyond the mylohyoid ridge from an anatomical viewpoint.

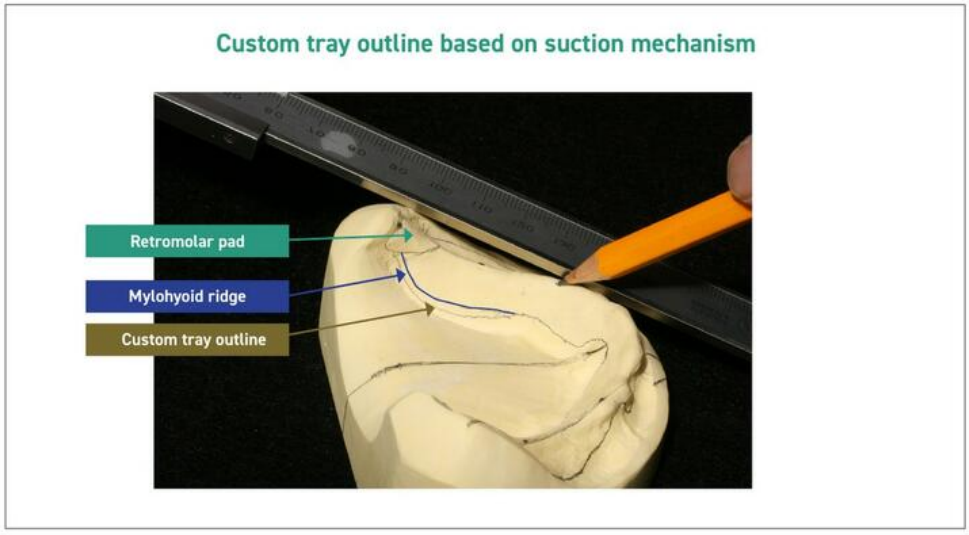



**Fig 4-22** A wall is built to resist tongue pressure as a convenient measure for creating a seal (top to bottom).

## 3) Sealing mechanism in the retromolar pad region

It is of utmost importance for readers seeking mandibular denture suction to understand the sealing mechanism in the retromolar pad region (**Fig 4-23a**). It consists of the close-contact closure formed between the mucosa in the retromolar pad region and the interior denture base surface, what the author calls the "interior seal" (**Fig 4-23b** area outlined with a dotted white circle), and close-contact closure formed between the side of the tongue and buccal mucosa over the denture base in the retromolar pad region termed the "exterior seal"(**Fig 4-23c** blue arrow). The former provides the main sealing function since the contact between the tongue and cheek (exterior seal) is lost during mouth opening (**Fig 4-23e**). The latter seals the denture from outside and above the retromolar pad when the upper and lower teeth come together during swallowing, which facilitates a negative pressure build-up. This is regarded as a secondary seal (**Fig 4-23f**). Both types of closure are required to ensure a clinically stable seal. Without an adequate understanding of the suction-effective denture seal mechanism, the area surrounding the retromolar pad is the most challenging area to achieve suction.



**Fig 4-23a** Seal in the retromolar pad region.

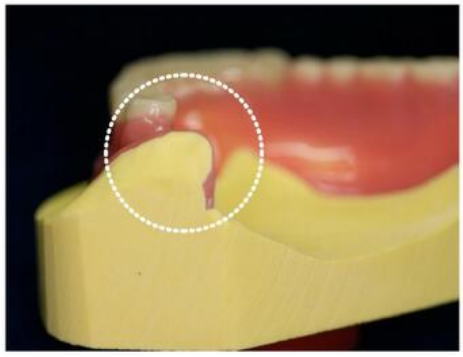


**Fig 4-23b** Closure with the contact of the tongue and buccal mucosa above the denture base in the retromolar pad region.

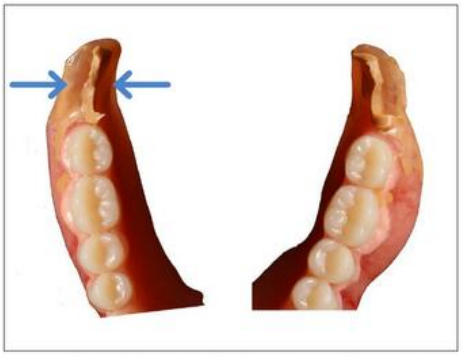


**Fig 4-23c** Close contact closure of the interior denture base surface.

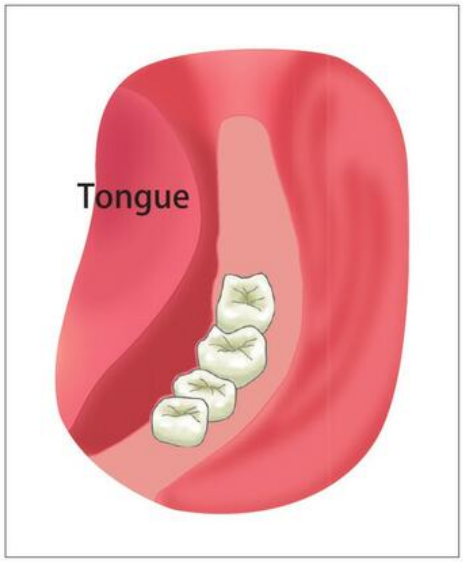



**Fig 4-23d** Denture placed in the opend mouth position.

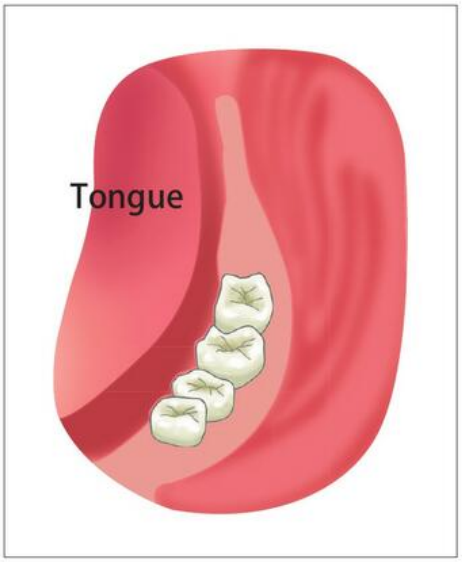



**Fig 4-23e** Importance of interior seal in half opened mouth position (main seal).

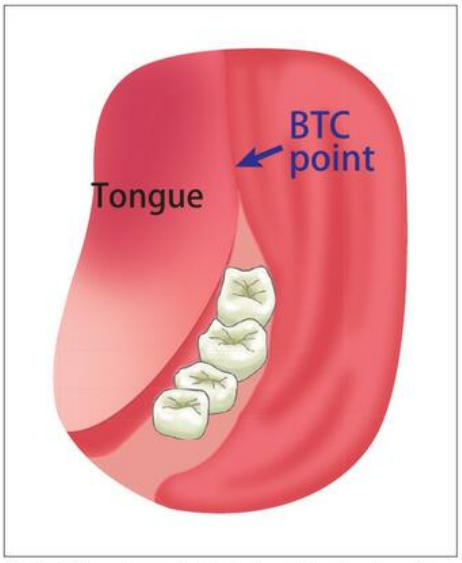



**Fig 4-23f** Secondary role of exterior seal in closed mouth position.

### 1 Close contact closure with the mucosal surface in the retromolar pad region and the interior surface of the denture base

The sealing mechanism in this region is essentially the same "close-contact closure" as in the maxillary posterior palatal area (**Fig 4-24**). However, extra care must be taken in this region because of the significantly smaller contact area when compared to the palate.

Many muscles are attached around the retromolar pad, but there is no direct muscle attachment to the retromolar pad itself. The retromolar pad deforms not only when pressure is applied during an impression but is also stretched by the pterygomandibular fold during mouth opening, which pulls the posterior border of the retromolar pad up and back. The retromolar pad is thus constantly exposed to the risk of deformation.

How should an impression of the retromolar pad be taken, and in what state or shape should the retromolar pad be recorded? When a mandibular edentulous impression is taken, should the mouth be open or closed, and should the clinician apply any pressure? Please consider the following answer. Negative pressure builds up on the interior surface of the denture base the moment the denture sinks into the denture-bearing mucosa. The target shape of the retromolar pads to be captured in both the preliminary and final impressions should be the natural form of the retromolar pads when the mouth is closed and the mandible is relaxed. In summary, the process of achieving mandibular denture suction starts with the impression of the static shape of the oral cavity including the retromolar pads when the mandible is at rest (**Fig 4-25**). The Frame Cut Back Tray (FCB Tray) (**Fig 4-26**)was designed to achieve this objective as a tray for the preliminary impression of the completely edentulous mandible. The sections of the FCB tray that cover the distal parts of the retromolar pads and the buccal shelves have been removed to avoid deformation of the retromolar pads and surrounding areas in the impression. A well-fitting custom tray can then be constructed on the cast poured into the preliminary impression taken with this tray, providing clearly defined anatomical structures to establish close-contact closure in the retromolar pad region. Features of the FCB Tray and details on how to use the tray will be described in the "Mandibular Preliminary Impression" section on Chapter 5 P.90.

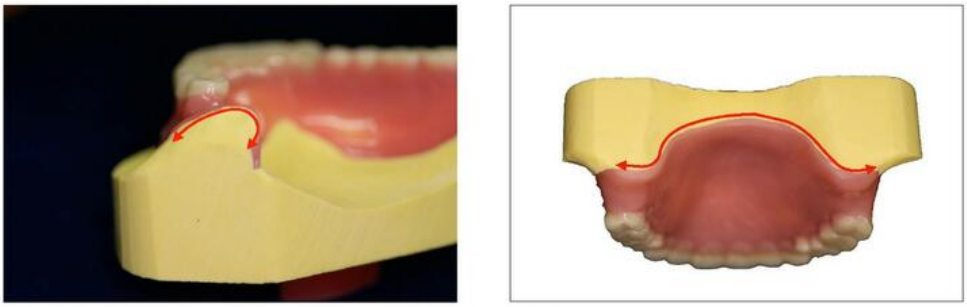


**Fig 4-24a, b** Close contact closure for the mandibular and maxillary dentures.



**Fig 4-25a, b** Close contact closure in the retromolar pad region. The interior surface of the denture base is in close contact with the mucosa above the retromolar pad.



**Fig 4-26** Frame Cut Back Tray (manufactured by YDM. CORPORATION, marketed by J. Morita Corporation).

## 2 Posterior border seal with the tongue and buccal mucosa above the polished surface in the retromolar pad region

### a. BTC Point

The posterior border seal of the mandibular complete denture is created by the intimate contact between the buccal mucosa and tissues of the tongue that meet above the retromolar pad. In 1988, Onoki [13] used an endoscope to demonstrate that the buccal mucosa lay over the denture base in the retromolar pad region as the mouth closed (**Fig 4-27**). The author later stressed that a posterior border seal was reinforced by the contact of the buccal mucosa with the side of the tongue above the retromolar pad when the mouth was closed, and named this contact the "BTC Point" (Buccal mucosa and Tongue side Contact Point) (**Fig 4-28**).

### b. Formation of the BTC Point

**Fig 4-29** depicts the location of the BTC Point in relation to the posterior border seal. The left panel of **Fig 4-29** shows that the BTC Point is generally located in the distal 1/3 of the retromolar pad, which means that the close-contact closure formed by the buccal mucosa and tongue above the retromolar pad will not be complete unless the denture base extends past the BTC Point.

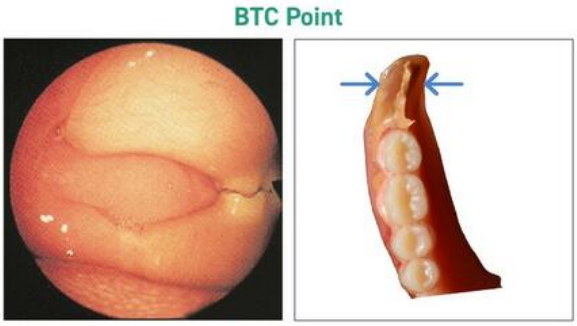
The right panel of **Fig 4-29** illustrates a denture base without the BTC Point, causing the seal to be broken due to air leakage. This demonstrates that the posterior border seal will be incomplete unless the entire retromolar pad is covered with a thin layer of denture base. On the other hand, if the denture base is too thick in this area, it will interfere with the formation of the BTC Point and the posterior border seal will be incomplete.

There have been discussions how much of the retromolar pad should be covered with the denture base (**Table 4-1**). Some researchers consider the retromolar pad mainly as a pressure-resistant tissue and insist that this objective could be accomplished by covering approximately the anterior half of the retromolar pad which is rich in fibrous tissue. However, when the main objective is to achieve suction that requires a peripheral seal around the entire denture base, as suggested by the author, it is essential to cover the entire retromolar pad. Even a single spot of air leakage makes the seal incomplete, resulting in a loss of suction. As a result, the posterior denture border not only covers the entire retromolar pad but also contacts the soft glandular tissue posterior to the retromolar pad. In summary, the entire denture border needs to be in contact with the soft tissue to create a strong seal, as mentioned in the short summary of "Suction Mechanism of the Maxillary and Mandibular Complete Dentures" (see P.42).

In comparison, when a maxillary denture covers the entire maxillary palate, its posterior border contacts the soft palate which results in a strong retentive seal. On the other hand, a palate-less denture that has been significantly reduced and only partly contacts the hard palate, has a markedly weakened suction seal. As a denture moves on the ridge mucosa while in function, the border seal can be more effectively maintained against this denture movement if the entire border of the denture base is in contact with the soft tissues.

**Fig 4-27** Onogi reported in 1988 that the buccal mucosa lies over the retromolar pad in closed-mouth position.

**Fig 4-28** BTC Point: Posterior border seal with the buccal mucosa contacting the side of the tongue over the distal part of the retromolar pad in closed-mouth position.



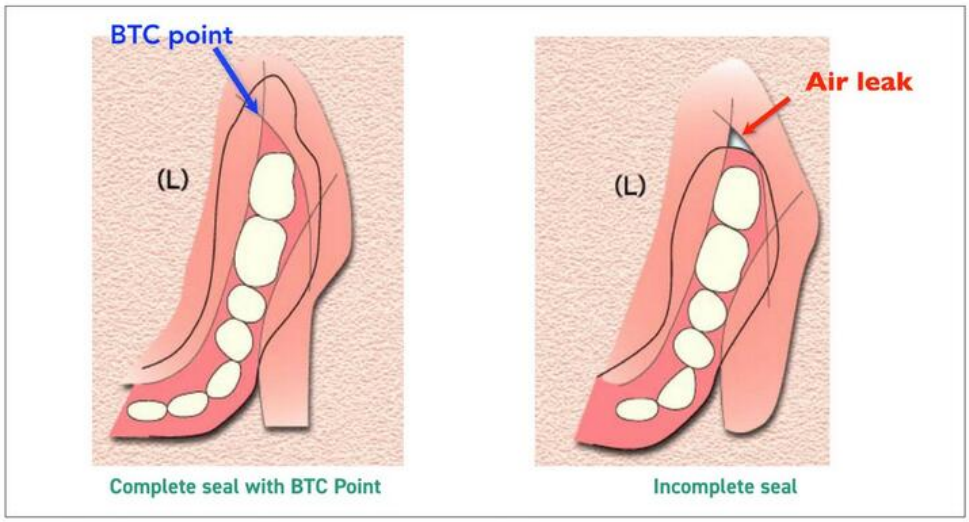




**Fig 4-29** Complete seal (suction-effective) and incomplete seal (suction-ineffective).

| Full coverage of retromolar pad | Partial coverage of retromolar pad |
|---|---|
| Schreinemakers J. [14] | Zarb, et al.: 1/2-1/3 [18] |
| Barnard Levin [15] | Masakata Yasaki: Anterior border [19] |
| Hiroshi Muraoka [16] | Keizo Mine: 2/3 [20] |
| Seiichiro Someya [17] | Iwao Hayakawa [21] |
| | Shizuo Toyoda:1/2-1/3 [22] |
| | Jun Nishiura: 1/2 or more [23] |

**Table 4-1** Previously reported degrees of retromolar pad coverage.

## c. Reasons why the BTC Point is easier to obtain in the mandible

Why is the BTC Point more easily formed in the posterior mandible? The reasons are threefold:

1. The movable volume of the mucobuccal fold is greater in the mandible than in the maxilla.
2. The mandibular alveolar ridge is lingually inclined, making it easier for the buccal mucosa to move inward. In contrast, the maxillary alveolar ridge is more upright and keeps the buccal mucosa from moving in (**Fig 4-30**).
3. A frenum-like structure, called the "Someya's sinew string," is present at the buccal root of the retromolar pad, and it strongly pulls the buccal mucosa inward.

What is Someya's sinew string?

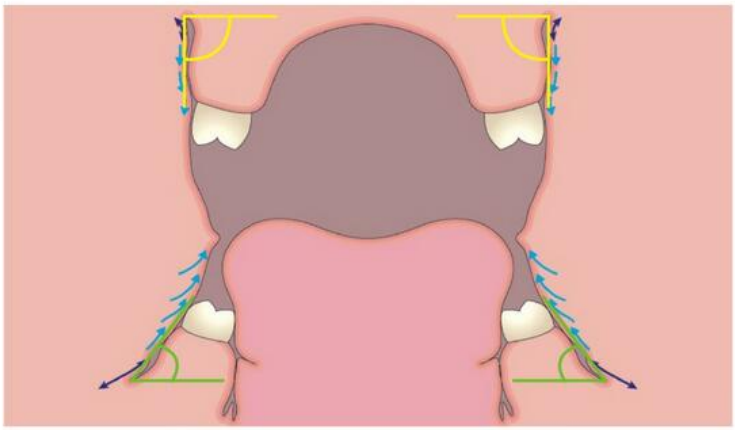
Under tension, a mucosal sinew-like string is formed at the buccal root of the retromolar pad posterior to the second molar. Dr. Seiichiro Someya first discovered and reported this structure; it is named after him and thus called the "Someya's sinew string" (**Fig 4-31**) [21]. Since the presence or role of this sinew string has not been demonstrated clearly from a prosthodontic or anatomical perspective, the Someya's sinew string is yet to be recognized as an official anatomical dental term.

It is thought that the action of the sinew string is to pull the buccal mucosa strongly inward during swallowing and close the space posterior to the second molar, contributing to the formation of the BTC Point. However, edentulous individuals can exert only about 1/6 of the occlusal force of dentate ones, and the sinew string tends to become less pronounced as the occlusal force decreases. According to Someya [17], the sinew string was visible in only 10 to 20% of the subjects he studied. However, the rate increased to approximately 40% by placing a dental mirror or a finger against the area buccal to the retromolar pad and spreading the buccal mucosa outwards. An electron microscopic view of the sinew string in an edentulous patient shows a fibrous tissue rather than a ligamentous or frenum-like tissue (**Fig 4-32**).

## d. Space allowing the formation of the BTC Point

The creation of the BTC Point is a natural physiological function that every human being, edentulous or dentulous, possesses. The acellular tendons are identified on the slice specimen(**Fig 4-32**). The fact that food does not get trapped in the space posterior to the retromolar pad seems to prove the presence of the BTC Point[35]. To allow the formation of the BTC Point, there must be a space above the retromolar pad for the buccal mucosa and tongue to enter. The size of this space between the retromolar pad and maxillary tuberosity should be at least 3 mm. Healthy dentate individuals are said to have between 5.0 to 7.0 mm of this space, indicating that a minimum of 3 mm of space is available for the formation of BTC Point in edentulous patients, provided a proper occlusal vertical dimension can be established (**Fig 4-33**). If the patient's occlusal vertical dimension cannot be raised for one reason or another, the space required for the formation of BTC Point cannot be secured. This makes it difficult to fully cover the retromolar pad with the denture base, thus eliminating the possibility of achieving denture suction.

**Fig 4-30** A large movable volume of the mandibular mucobuccal fold and the lingual inclination of alveolar ridge are conducive to formation of the BTC Point.



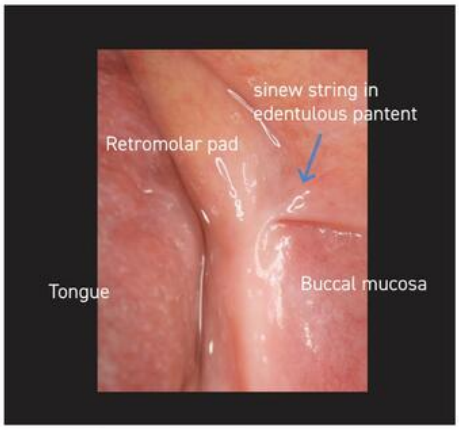



**Fig 4-31a** Someya's sinew string serves to pull the buccal mucosa inward.




**Fig 4-31b** In order not to interrupt the buccal mucosa to completely cover the retromolar pad region, slightly curvy, slightly deeper and even deeper notch of the weak, moderate, strong sinew string should be created respectively.



**Fig 4-32** SEM image of Someya's sinew string rich in collagen (courtesy of Department of Anatomy, Tokyo Dental College).




**Fig 4-33** At least 3 mm of space is required for the formation of the BTC Point above the denture base in the retromolar pad region.

### e. How is the BTC Point formed? (Observations with an endoscopic video and MRI)

An endoscopic video was used to observe the formation of the BTC Point. A duplicate denture was fabricated, and a hole was drilled in the anterior to posterior direction, as shown in **Fig 4-34**. The objective was to observe how the retromolar pad region is sealed when the mouth is opened and closed.

1) From open to closed-mouth position: The buccal mucosa extends over the denture base in the retromolar pad region. The tongue moves towards the retromolar pad to come into contact with the buccal mucosa, resulting in the formation of the BTC Point. The same phenomenon was observed with an MRI.
2) From closed to open-mouth position: The BTC Point is lost as the mouth opens. The buccal mucosa moves out towards the buccal side of the retromolar pad, while the tongue moves inward and downward. Thus, the contact closure above the retromolar pad is lost. The stability of the denture was maintained only with the close contact closure between the mucosa lining the retromolar pad and the interior surface of the denture base (**Fig 4-35**).

From these observations, the main player of the seal in the retromolar pad region is found to be the close-contact closure created by the lining mucosa of the retromolar pad and the interior surface of the denture base, which prevents the denture from lifting even when the BTC Point is lost during mouth opening. These findings also indicate that the formation of the BTC Point above the retromolar pad is necessary to ensure the build-up of negative pressure on the interior surface of the denture base during mouth closure.




**Fig 4-34** Endoscopic and MRI observations of BTC Point.

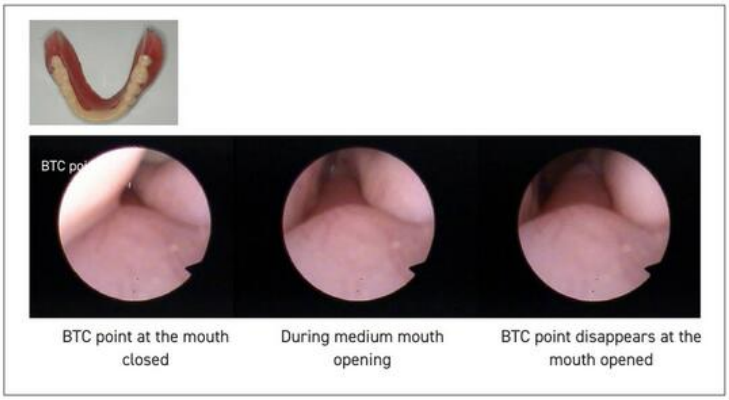



**Fig 4-35**

### f. A study on factors interfering with the formation of the BTC Point [27]

First, the normal condition with the healthy formation of BTC points in edentulous are shown(**Fig 4-36**). A study was conducted to investigate the effects of buccal and lingual overextensions of the denture base on the formation of the BTC Point (posterior seal) by extending the denture base with a utility wax in the buccal part of the retromolar pad, the lateral part of the buccal shelf, and the lingual part into the retromylohyoid fossa region. An endoscope was used to observe the movement of the buccal mucosa and tongue when the mouth was opened and closed.

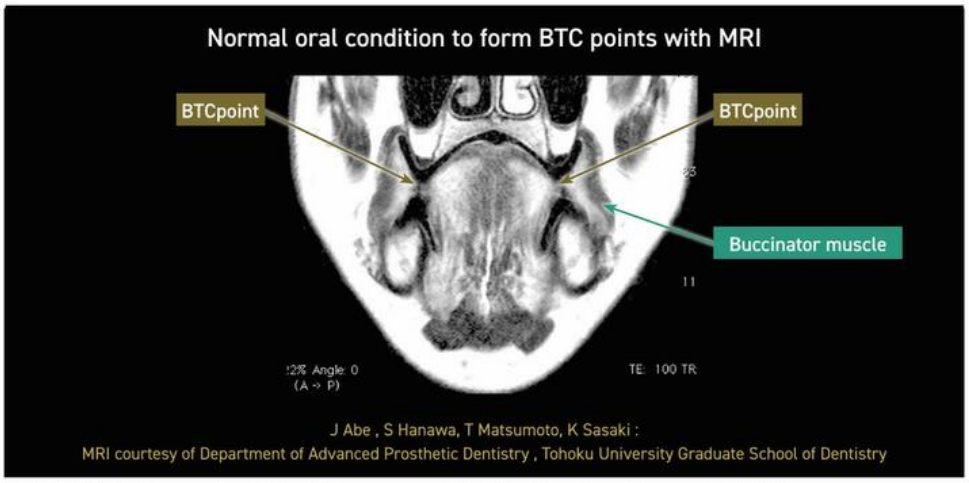



**Fig 4-36** MRI also demonstrates that the utillity wax prevents BTC Point formation on the overextended side.

## 3 Notice! not to break the seal

### [1] When the denture base was overextended buccally with a utility wax:

The utility wax prevented the buccal mucosa from covering the retromolar pad. Thus, the BTC Point was not formed. This finding was confirmed with MRI (**Fig 4-37**).

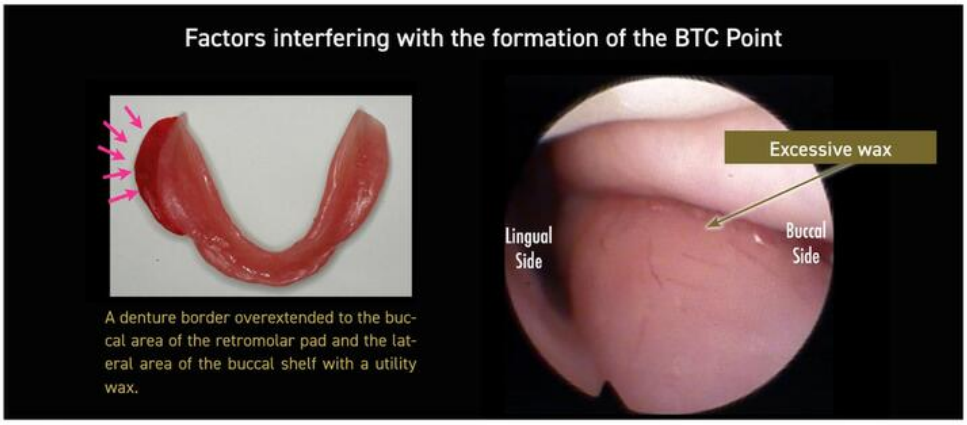



**Fig 4-37a** The moment of the buccal mucosa to cover the retromolar pad was distributed by the utility wax, preventing BTC point formation.

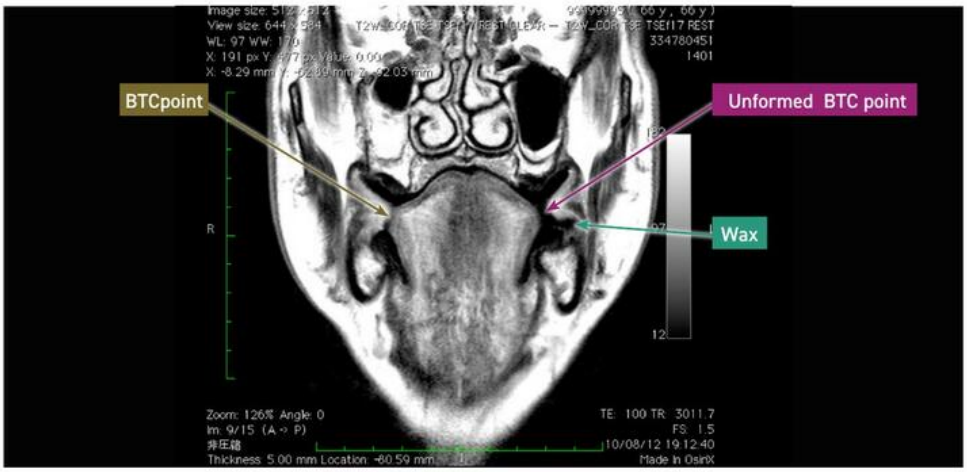



**Fig 4-37b** MRI also demonstrates that the utility wax prevents BTC Point formation on the overextended side.

## [2] When the denture base was overextended not only buccally but lingually into the retromylohyoid fossa region with a utility wax:

The tongue was constrained by the utility wax and prevented from moving up to the level of the retromolar pad even during mouth closure (**Fig 4-38**).

This study clarified that it is important not to disturb the movement of the buccal mucosa hanging over the retromolar pad during mouth closure in order to ensure the formation of the BTC Point over the posterior border of the denture base.

The dentist may try to extend the denture base into the buccal shelf area using a compound material during the final impression with the aim of increasing a denture-supporting area. In this case, it is possible to extend the denture base beyond the insertion of the buccinator muscle comprised of horizontal muscle fibers. It is also possible in some cases to extend the denture base by 1 cm down towards the inferior part of the retromylohyoid fossa region. However, if these attempts to increase a denture-supporting area end up hindering the formation of the BTC Point, the fundamental physiologic function of "posterior seal" would be lost. The patient would then perceive such a denture as an ill-fitting, cumbersome denture.

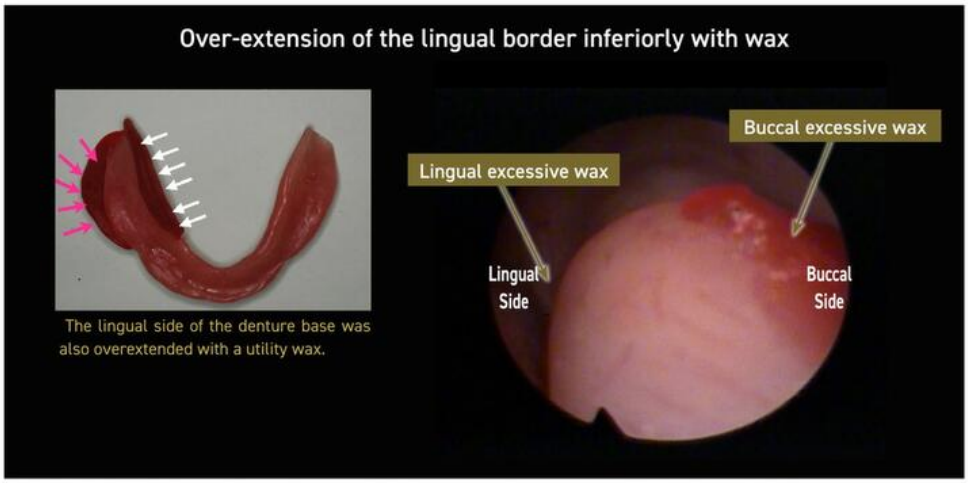



**Fig 4-38** The BTC Point was not formed because of interference with tongue movement when the mouth is closed.

It is important to mark the custom tray outline along the mucosal fold on the buccal side of the retromolar pad in such a way that the tray will not disturb the action of the buccal mucosa to cover the retromolar pad (**Fig 4-39**). If the outline extends beyond this mucosal fold, the tray would disturb BTC Point formation by preventing the buccal mucosa from covering the retromolar pad (**Fig 4-39**). The impression technique using a FCB Tray that minimizes deformation of the retromolar pad will capture the exact anatomical location of the mucosal fold in the impression.

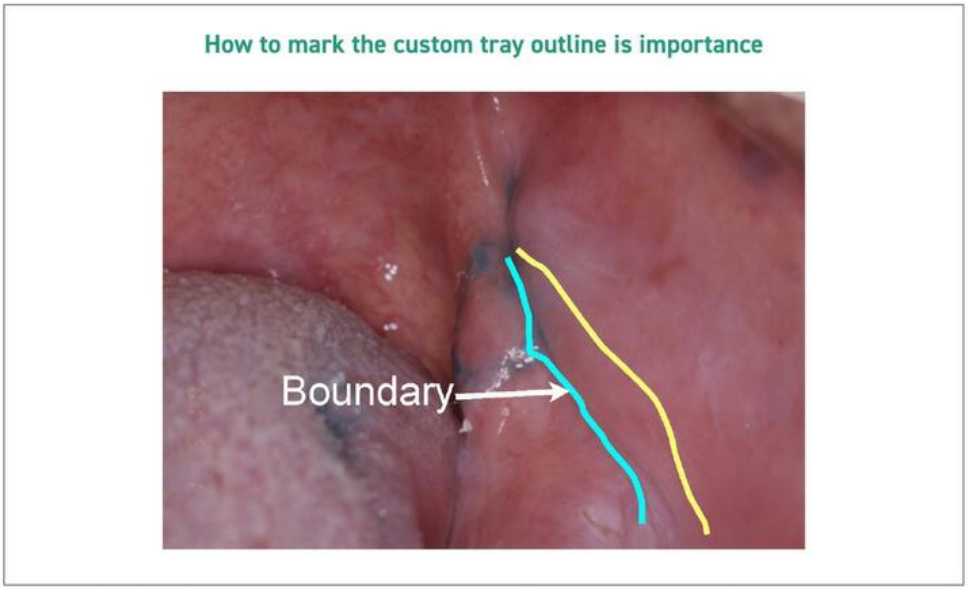



**Fig 4-39** The junction between retromolar pad and buccal mucosa must be accurately captured to increase the accuracy of fit of the custom tray in the retromolar pad region. Blue line : Adequate custom tray design at the buccal shelf region. Yellow line : overextended custom tray design affecting the denture border seal.

## Short Summary

The suction mechanism of the mandibular complete denture differs among different areas of the mandibular ridge and therefore is more complex than in the maxilla. It consists of

1) labial and buccal interior/exterior double closure (blue),
2) lingual interior/exterior double closure with spongy tissue in the sublingual fold area (yellow),
3) compensatory closure in the retromylohyoid fossa region (green), and
4) contact closure created by the mucosa lining the retromolar pad and the interior surface of the denture base (interior seal), and contact closure created by the side of the tongue and the buccal mucosa above the retromolar pad (exterior seal) (red) (**Fig 4-40**).

It is import for both dentists and dental technicians involved in denture fabrication to fully understand the suction mechanism, which all boils down to the basic principle that denture seal will not be complete when there is even a single place of air leak.

**Summary of sealing mechanism of the mandibular complete denture**

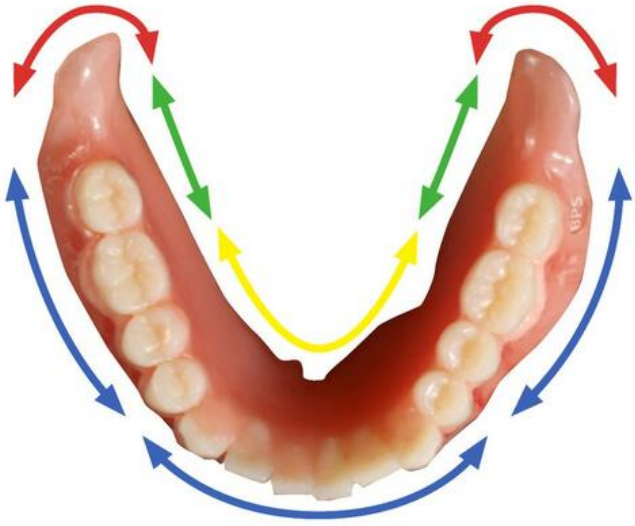


Fig 4-40